

Prabhas K Deshpande

ANALOG & MIXED-SIGNAL IC DESIGNER

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PROFESSIONAL SUMMARY

Analog & Mixed-Signal IC Designer with industry and research experience in circuit design, analysis, and silicon implementation, with a strong interest in developing high-performance and robust analog circuits. My experience spans circuit-level design, simulation, design optimization, and practical implementation, providing an understanding of the challenges involved in translating circuit concepts into working silicon. I am looking to contribute to a technically rigorous and collaborative team where I can take ownership of challenging design problems, continuously deepen my expertise, and contribute meaningful solutions to real-world engineering challenges.

MS THESIS & RESEARCH ACTIVITIES

- 📖 **Thesis:** A High Input Impedance Chopper Stabilized Amplifier For Interfacing Dry Electrodes (Preprint available on arXiv: [2606.13129](#)). Proposed an innovative input impedance (Z_{in}) boosting architecture optimized for bio-potential signal acquisition with Dry Electrodes, featuring an intelligent capacitor reconfiguration strategy within the signal chain.
- 🌟 **Patent:** “A High Input Impedance Chopper Stabilized Amplifier” – Indian Patent Application No. **202641027947** (Filed, March 2026).
- 🏢 **PDK Experience:** Hands-on circuit design and physical verification familiarity across TSMC 65nm, SCL 180nm, and IHP 130nm.
- ⚙️ **FOSS CAD:** Explored open-source CAD toolchains (Xschem, NGspice, Klayout, Magic) for a global collaborative tape-out framework organized under the IEEE UNICASS initiative (Tapeout expected in October 2026).

EDUCATION

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|---|------------------------------------|
| 🎓 Master of Science (by Research) in Electrical Engineering
<i>Indian Institute of Technology Dharwad</i> | 2024 – Ongoing
8.95 CGPA |
| 🎓 Bachelor of Engineering in Electronics & Communication
<i>KLE Technological University</i> | 2018 – 2022
8.64 CGPA |

EXPERIENCE

Analog Layout Engineer | Texas Instruments 2022 – 2024

- Translated critical analog chain blocks (OPAMPs, bias generators, DACs, level shifters) into silicon, establishing a deep, layout-aware understanding of device matching, parasitic mitigation, and physical non-idealities.
- Worked on Design automation scripts that enabled accelerating workflows to achieve faster convergence.

Member | Research Scholars Forum (RSF), EECE Dept, IIT Dharwad 2025 – Ongoing

- Co-pioneered the department's first autonomous student body to cultivate a dynamic academic environment; co-organized **Vivriti-2026**, the annual research symposium.

TECHNICAL SKILLS

 **EDA Tools & Platforms:** Cadence Virtuoso, Assura, Calibre, Xschem, Klayout, Magic, KiCad

 **Simulators:** Spectre, NGspice

 **Languages:** Cadence-SKILL, Python, Verilog

COURSE WORK (MASTERS)

 Analog IC Design, Digital IC Design, Mixed-Signal IC Design, Embedded Systems Design, Linear Algebra & Its Applications.

HOBBY PROJECT

★ **LLM-Driven Analog Netlist Framework:** Explored Large Language Models to automate netlist generation for analog circuits, aiming to accelerate early-stage design exploration.

INTERESTS / HOBBIES

★ **Music:** Self-taught amateur music producer; enjoy tinkering with FL Studio and playing noob-level guitar.  [YouTube](#).

★ **Recreation:** Play casual volleyball and cricket, and enjoy local hiking.

PROFESSIONAL REFERENCES

- **Dr. Naveen Kadayinti**
 - **Position:** Associate Professor, Dept. of Electrical, Electronics and Communication Engineering, IIT Dharwad
 - **Email:** naveen@iitdh.ac.in
 - **Relationship:** Guide and Academic/Research Advisor
- **Dr. Samatha Benedict**
 - **Position:** Assistant Professor, Dept. of Electrical, Electronics and Communication Engineering, IIT Dharwad
 - **Email:** samatha.benedict@iitdh.ac.in
 - **Relationship:** Guide and Academic/Research Advisor
- **Dr. Abhijit Kshirsagar**
 - **Position:** Associate Professor, Dept. of Electrical, Electronics and Communication Engineering, IIT Dharwad
 - **Email:** kabhijit@iitdh.ac.in
 - **Relationship:** Academic/Research Advisor